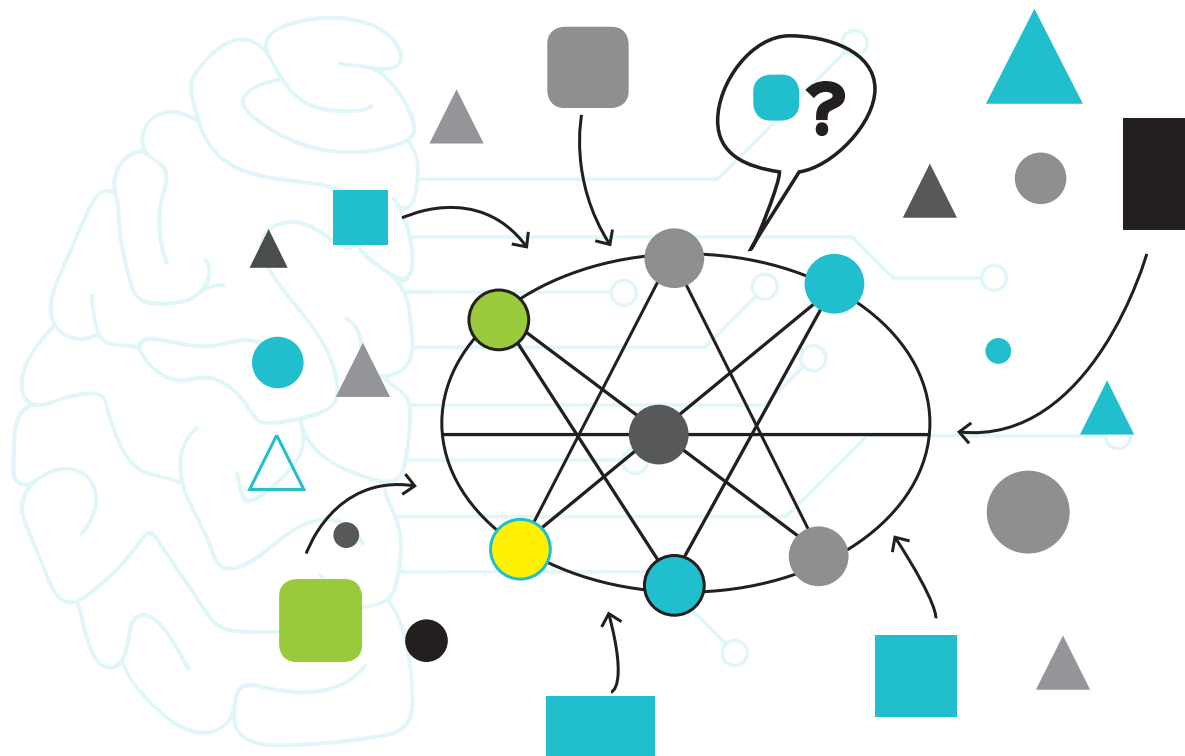


The Long and Short of Federated Machine Learning



This article gives a quick insight into what is federated machine learning and why it's popular today.

A cloud refers to a server, public or private, accessible over a network, which is usually the Internet. It generally has high processing power and storage, and is suitable for big computations. A cloud can be used to train AI models, but only when the data is available to it. On the other hand, since the cloud is generally a remote system, capturing data directly on it becomes difficult and not feasible. Capturing the data on local devices and transmitting it to the cloud does not always give real-time results. This is where the concept of federated learning comes in.

Federated learning promotes machine learning while the data is on the device. It handles a flexible architecture, which enables a secure process for sensitive data collection and model training. The world currently takes data privacy as an important responsibility. To introduce automation into fields like healthcare, biometrics, etc, real-time sensitive data is the core requirement. The important question therefore is: How do we train a model without collecting sensitive data from the users, storing it and using it for training? To define optimal training for a machine learning model, relevance of

data is an important aspect. Data is best when it comes directly from a source. But the permission for data collection becomes an issue.

With federated machine learning, model training can be centralised on a decentralised data feed. The model is trained on the source device, and the device configurations are measured to see if it is able to train the model or not. Data sources are selected based on how optimally they can provide the data. Once the model trains on the device, it sends the training results (not the data) to the server. In a similar fashion, training results from edge devices are